

Topic 1[Exception Handling]

Problem Statement:

In the following input i is the index of array ax[]. The program prints ax[i]. Then write catch block if i is out of range of ax[]. Write three catch blocks to fulfill the purpose

- i) a catch block receives the value of i
- ii) a catch block receives string "Out of Range Error"
- iii) a default catch() if above two catch block doesn't match

Code	Output
<pre>#include<bits/stdc++.h> using namespace std; #define ll long long int #define str string #define newl cout<<endl int main() { int i; int ax[5]={10,20,60,40,30}; cout<<"enter index: "; cin>>i; try { if(i<5&& i>=0) { throw ax[i]; } else if(i>4) { str s="Out of Range Error"; throw s; } else { char c='a'; throw c; } } catch(int x) { cout<<"ax["<<i<<"]="<<x<<endl; } catch(str s) { cout<<s; } catch(...) { cout<<"Invalid Input"; } }</pre>	i) enter index: 3 ax[3]=40 Process returned 0 (0x0) execution time : 17.382 s Press any key to continue.
	ii) enter index: 7 Out of Range Error Process returned 0 (0x0) execution time : 0.516 s Press any key to continue.
	iii) enter index: -1 Invalid Input Process returned 0 (0x0) execution time : 0.891 s Press any key to continue.

Topic 2 [class Template]:

Problem Statement: In the following class data members x and y are integers and the method Sum() adds x and y. However, we need to perform the sum of int+int, int+double, double+int and double+double. To achieve it, change the definition of x, y, setData() and Sum() accordingly.

<pre>class A{ private: int x; int y; public: void setData(int x,int y){ this->x=x; this->y=y; }</pre>	<pre>int Sum(){ int s; s=x+y; return s; }; int main(){ //write required statements to call SetData() & Sum() }</pre>
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Code	Output
<pre>#include <bits/stdc++.h> using namespace std; #define ll long long int #define str string #define newl cout << endl template<typename t1,typename t2> class A { private: t1 x; t2 y; public: void setData(t1 x, t2 y) { this->x = x; this->y = y; } auto Sum() { auto s=x+y; return s; } }; int main() { A<int,int>a1; a1.setData(5,3); // int +int cout<<a1.Sum(); newl; A<int,double>a2; a2.setData(5,3.25); // int + double cout<<a2.Sum(); newl; A<double,int>a3; a3.setData(6.55,4); // double + int cout<<a3.Sum(); newl; A<double,double>a4; a4.setData(6.25,5.66); // double + double cout<<a4.Sum(); newl; }</pre>	<pre>8 8.25 10.55 11.91 Process returned 0 (0x0) execution time : 0.031 s Press any key to continue.</pre>

Topic 3 [STL:Array class]

Problem statement: Declare a STL array object ax with 6 elements and do the following:

- i) Assign 10,60,30,70,20 to ax using single statement
- ii) Print third element of ax using at() function
- iii) Print first element of ax using front() function
- iv) Print last element of ax using back() function
- v) Fill the elements of ax using fill() function
- vi) Test whether ax is empty or not using empty() function
- vii) Print size of ax
- viii) Print maximum size of ax using max_size() function
- ix) Print address of first element of ax using begin() function
- x) Print address of last element of ax using end() function

Code	Output
<pre>#include<bits/stdc++.h> using namespace std; #define ll long long int #define yes_or_no cout<<"YES\n":cout<<"NO\n" #define newl cout<<endl void print(array<int,6>&ax) { for(int i=0;i<6;i++) { cout<<ax[i]<<" "; } newl; } int main() { array<int,6>ax={10,60,30,70,20}; // i print(ax); cout<<ax.at(2);newl; // ii cout<<ax.front();newl; // iii cout<<ax.back();newl; // iv ax.fill(50);print(ax); // v ax.empty()?yes_or_no: // vi cout<<ax.size();newl; // vii cout<<ax.max_size();newl; // viii cout<<ax.begin();newl; // ix cout<<ax.end();newl; // x }</pre>	i) 10 60 30 70 20 0 ii) 30 iii) 10 iv) 0 v) 50 50 50 50 50 50 vi) NO vii) 6 viii) 6 ix) 0x61fe00 x) 0x61fe18 Process returned 0 (0x0) execution time : 0.037 s Press any key to continue.

Topic 4[STL: pair class]

Problem statement: Define a pair class object px with int and string elements. Write statements to do the following

- i) Assign 10 to int and “Rajshahi” to px using make_pair() function
- ii) Print int data member by first
- iii) Print string data member by second
- iv) Modify first data member to 20 using get<>() function
- v) Declare another pair bx and assign values to bx and swap it with ax

Code	Output
<pre>#include<bits/stdc++.h> using namespace std; #define ll long long int #define str string #define newl cout<<endl int main() { pair<int,str>px; px=make_pair(10,"Rajshahi"); // i cout<<get<0>(px);newl; // ii cout<<get<1>(px);newl; // iii get<0>(px)=20;cout<<get<0>(px);newl; // iv pair<int,str>bx; swap(px,bx); // v cout<<get<0>(bx)<<" "<<get<1>(bx);newl; }</pre>	<pre>ii) 10 iii) Rajshahi iv) 20 v) 20 Rajshahi Process returned 0 (0x0) execution time : 0.031 s Press any key to continue.</pre>

Topic 5 [STL: tuple class]

Problem statement: Define a tuple class object tx with int.string and double elements. Write statements to do the following

- i) Assign <100,"Kamal",3.5> to tx using make_tuple() function
- ii) Print int data member by get() function
- iii) Print string data member by get() function
- iv) Print double data member by get() function
- v) Modify third data member to 3.7 using get<>() function
- vi) Declare another tuple bx and assign values to bx and swap it with ax

Code	Output
<pre>#include<bits/stdc++.h> using namespace std; #define ll long long int #define str string #define newl cout<<endl int main() { tuple<int,str,double>tx; tx=make_tuple(100,"Kamal",3.5); // i cout<<get<0>(tx);newl; // ii cout<<get<1>(tx);newl; // iii cout<<get<2>(tx);newl; // iv get<2>(tx)=3.7; cout<<get<2>(tx);newl; // v tuple<int,str,double>bx; swap(tx,bx); // vi cout<<get<0>(bx)<<" "; cout<<get<1>(bx)<<" "; cout<<get<2>(bx);newl; }</pre>	<pre>ii) 100 iii) Kamal iv) 3.5 v) 3.7 vi) 100 Kamal 3.7 Process returned 0 (0x0) execution time : 0.045 s Press any key to continue.</pre>

Topic 6 [STL: vector class]

Code	Output
<pre> #include<bits/stdc++.h> using namespace std; #define ll long long int #define str string #define newl cout<<endl void Display(vector<int>&A) { for(int i:A) { cout<<i<<" "; } newl; } int main() { vector<int>ax; int A[10]={5,9,4,7,6,8,10,15,10,14}; // i) for(int i=0;i<10;i++) { ax.push_back(A[i]); } Display(ax); // ii) ax.pop_back(); ax.pop_back();Display(ax); // iii) auto it=ax.begin(); advance(it,3); ax.insert(it,20);Display(ax); // iv) cout<<ax.front()<<endl; // v) cout<<ax.back()<<endl; // vi) cout<<ax.at(4)<<endl; // vii) auto i=ax.begin(); advance(i,6); ax.erase(i);Display(ax); // viii) cout<<ax.size()<<endl; // ix) cout<<ax.capacity()<<endl; // x) ax.reserve(5); cout<<ax.size()<<endl; // xi) ax.shrink_to_fit(); cout<<ax.capacity()<<endl; // xii) ax.resize(5); Display(ax); // xiii) ax.clear(); cout<<ax.size()<<endl; } </pre>	<pre> i) 5 9 4 7 6 8 10 15 10 14 ii) 5 9 4 7 6 8 10 15 iii) 5 9 4 20 7 6 8 10 15 iv) 5 v) 15 vi) 7 vii) 5 9 4 20 7 6 10 15 viii) 8 ix) 16 x) 8 xi) 8 xii) 5 9 4 20 7 xiii) 0 </pre> Process returned 0 (0x0) execution time : 0.040 s Press any key to continue.

Topic 7 [STL: stack class]

Problem statement: Write a program to create and manipulate Stack using stack class and Perform the following operations using the specified method.

- i) use push() method to push data
- ii) use pop() method to pop data
- iii) use top() method to display top element
- iv) use empty() method to check whether stack is empty or not

Code	Output
<pre>#include<bits/stdc++.h> using namespace std; #define ll long long int #define str string #define yes_or_no cout<<"YES\n":cout<<"NO\n" #define newl cout<<endl int main() { stack<int>st; st.push(10); st.push(20); // i cout<<st.top();newl; st.pop(); // ii cout<<st.top();newl; // iii st.empty()?yes_or_no; // iv }</pre>	i) 20 iii) 10 iv) NO Process returned 0 (0x0) execution time : 0.042 s Press any key to continue.

Topic 8 [STL: queue class]

Problem statement: Write a program to create and manipulate Queue using queue class. Perform the following operations using the specified method.

- i) use push() method to push data
- ii) use pop() method to pop data
- iii) use front() method to display front element
- iv) use back() method to display rear element
- v) use empty() method to check whether queue is empty or not

Code	Output
<pre>#include<bits/stdc++.h> using namespace std; #define ll long long int #define str string #define yes_or_no cout<<"YES\n":cout<<"NO\n" #define newl cout<<endl int main() { queue<int>qu; qu.push(10); qu.push(20); qu.push(30); // i cout<<qu.front();newl; qu.pop(); // ii cout<<qu.front();newl; // iii cout<<qu.back();newl; // iv qu.empty()?yes_or_no; // v }</pre>	i) 10 iii) 20 iv) 30 v) NO Process returned 0 (0x0) execution time : 0.099 s Press any key to continue.

Topic 9 [STL: list class]

Problem statement: Write a program to create and manipulate linked list using list class and its following methods to

- i) insert 8 integers using **push_back()** method
- ii) insert two elements using **push_front()** method
- iii) display all the elements of the list in forward direction with user-defined Display() method using **begin()** and **end()** methods and iterator
- iv) display all the elements of the list in reverse direction with a user-defined DisplayRev() method using **rbegin()** and **rend()** method and iterator
- v) display front element using **front()** method
- vi) display back element using **back()** method
- vii) delete front element using **pop_back()** method
- viii) delete front element using **pop_front()** method
- ix) search an element **x** using **find()** method
- x) insert a new element **x** before an existing element **y** using **insert()** method
- xi) insert a new element **x** after an existing element **y** using **insert()** method
- xii) count a particular element **x**
- xiii) count a elements with condition using predicate function
- xiv) delete a particular element **x** with **erase()** method
- xv) delete first 4 elements with **erase()** method
- xvi) delete a particular element **x** with **remove()** method
- xvii) delete a elements with condition using **remove_if()** method using predicate function
- xviii) assign elements from another list using **assign()** method
- xix) assign elements from an array using **assign()** method
- xx) sort the list using **sort()** method
- xxi) Delete consecutive similar elements using **unique()** method

Code	Output
<pre>#include<bits/stdc++.h> using namespace std; #define ll long long int #define str string #define ev(x) x%2==0 #define od(x) x%2==1 #define yes_or_no cout<<"YES":cout<<"NO" #define newl cout<<endl void Display(list<int>&li) { auto i=li.begin(); auto j=li.end(); while(i!=j) { cout<<*i<<" "; i++; } } void DisplayRev(list<int>&li) { auto i=li.rbegin(),j=li.rend(); while(i!=j) { cout<<*i<<" "; i++; } } int Even_count(list<int>&li) { int x=0; for(auto i:li)</pre>	

<pre> { if(ev(i)) { x++; } } return x; } bool is_odd(int x) { return od(x); } int main() { int A[]={10,64,25,36,78,64,69,70,64,67}; list<int>li; for(int i=1;i<=8;i++) { li.push_back(A[i-1]); } // i) li.push_front(A[8]); li.push_front(A[9]); // ii) Display(li);newl; // iii) DisplayRev(li);newl; // iv) cout<<li.front();newl; // v) cout<<li.back();newl; // vi) li.pop_front();Display(li);newl; // vii) li.pop_back();Display(li);newl; // viii) find(li.begin(),li.end(),78)!=li.end()?yes_or_no;cout<<" "; find(li.begin(),li.end(),100)!=li.end()?yes_or_no;newl; // ix) auto x=li.begin();advance(x,2); li.insert(x,100);Display(li);newl; // after 10 // x) auto y=li.begin();advance(y,4); li.insert(y,50);Display(li);newl; // before 25 // xi) cout<<count(li.begin(),li.end(),64);newl; // xii) cout<<Even_count(li);newl; // xiii) auto z=li.begin();advance(z,9); li.erase(z);Display(li);newl; // xiv) for(int i=1;i<=4;i++) { li.erase(li.begin()); } Display(li);newl; // xv) auto b=remove(li.begin(),li.end(),78); li.erase(b,li.end()); Display(li);newl; // xvi) auto a=remove_if(li.begin(),li.end(),is_odd); li.erase(a,li.end()); Display(li);newl; // xvii) list<int>B={15,25,51,45,60,80,64}; li.assign(B.begin(),B.end()); Display(li);newl; // xviii) li.assign(begin(A),end(A)); Display(li);newl; // xix) li.sort();Display(li);newl; // xx) auto it=unique(li.begin(),li.end()); li.erase(it,li.end()); Display(li);newl; // xxi) } </pre>	<pre> iii) 67 64 10 64 25 36 78 64 69 70 iv) 70 69 64 78 36 25 64 10 64 67 v) 67 vi) 70 vii) 64 10 64 25 36 78 64 69 70 viii) 64 10 64 25 36 78 64 69 ix) YES NO x) 64 10 100 64 25 36 78 64 69 xi) 64 10 100 64 50 25 36 78 64 69 xii) 3 xiii) 8 xiv) 64 10 100 64 50 25 36 78 64 xv) 50 25 36 78 64 xvi) 50 25 36 64 xvii) 50 36 64 xviii) 15 25 51 45 60 80 64 xix) 10 64 25 36 78 64 69 70 64 67 xx) 10 25 36 64 64 64 67 69 70 78 xxi) 10 25 36 64 67 69 70 78 Process returned 0 (0x0) execution time : 0.082 s Press any key to continue. </pre>
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